

Tea Leaf Disease RAG Agent Knowledge Base

Agent-optimized PDF for disease classification + pixel severity + advisory response generation

Purpose: This document is written for retrieval-augmented generation (RAG). It is not a textbook chapter. Each section uses repeated disease names, aliases, symptoms, severity rules, and action blocks so that a retriever can find the correct advice from predicted disease class and severity level.

System input fields expected: predicted_disease, severity_percent, severity_level, confidence_score, optional_location, optional_user_question.

Class labels used by the model: algal_leaf_spot, brown_blight, gray_blight, healthy, helopeltis.

Severity thresholds used in this project: healthy = 0%; mild = >0-15%; moderate = >15-40%; severe = >40%.

Safety and Scope Rules for the RAG Agent

- **Always ground advice in the disease and severity given by the model.** If the prediction confidence is low, recommend expert verification before treatment.
- **Do not recommend banned or unregistered pesticides.** The agent should say: use locally registered products and follow the label, tea board/BTRI guidance, pre-harvest intervals, and safety requirements.
- **Prioritize IPM:** monitoring, sanitation, pruning, drainage, shade management, removal of infected leaves/shoots, and biological/botanical options where suitable.
- **Chemical advice should be general and label-based.** Avoid unsafe mixing instructions. Mention active ingredient groups only when appropriate and advise local expert confirmation.
- **For severe cases:** advise consultation with an agricultural extension officer, tea garden agronomist, or BTRI/local plant protection authority.

Agent Response Template

Use this structure for final RAG responses:

Disease: {predicted_disease} Severity: {severity_level} ({severity_percent}%) Likely issue: {short explanation} Recommended action: {severity-specific action} Monitoring: {when to recheck} Safety note: Follow local label/BTRI/extension guidance, PPE, and pre-harvest interval. Confidence note: If model confidence is low or symptoms do not match, confirm with an expert.

Severity-Based General Action Rules

Severity	Pixel severity range	General agent action
Healthy	0%	No treatment. Recommend routine monitoring, good drainage, balanced nutrition, sanitation, and disease-preventive field hygiene.
Mild	>0-15%	Early intervention. Remove affected leaves/shoots where practical, improve airflow, monitor weekly, and use preventive registered products only if risk is increasing.
Moderate	>15-40%	Active intervention. Combine sanitation, pruning/airflow/drainage correction, and locally registered treatment. Recheck after 7-14 days.
Severe	>40%	Urgent action. Isolate/mark affected area, remove heavily affected material, consult expert/extension service, and apply registered treatment following label and harvest interval.

Agent Card: brown_blight

Field	Value
Canonical class label	brown_blight
Human-readable name	Brown blight
Problem type	Fungal leaf disease
Retrieval aliases	brown blight, tea brown blight, leaf blight, Colletotrichum blight
Best query keywords	brown_blight moderate treatment; brown blight severe tea; brown_blight mild advice; fungal leaf blight tea treatment

Recognition clues for the agent

- Brown irregular patches from leaf edge or tip
- Necrotic dry tissue with clear boundary
- Lesions may enlarge inward toward midrib
- Often favored by warm, humid, wet conditions

Short explanation to include in answers

The model identified brown blight-like symptoms. Severity is calculated from predicted infected pixels divided by total leaf area.

Severity-specific action blocks

Severity level	Agent treatment/advice block
mild (>0-15%)	Remove visible infected leaves during routine field sanitation Improve air circulation and avoid prolonged leaf wetness Monitor nearby leaves weekly If disease pressure is high, consider a preventive locally registered copper/contact fungicide according to label
moderate (>15-40%)	Remove and destroy infected leaf material Improve drainage, shade balance, and pruning to reduce humidity Use a locally registered fungicide program appropriate for tea, rotating modes of action Recheck after 7-14 days and record spread
severe (>40%)	Consult tea extension/agronomist/BTRI for confirmation Prune heavily affected shoots or sections if needed Use registered contact/systemic fungicide strategy according to label and pre-harvest interval Increase monitoring frequency until new growth is healthy

Do-not-recommend / safety guardrails

- Do not overuse nitrogen fertilizer
- Do not spray during rain or without PPE
- Do not reuse contaminated pruning tools without cleaning

RAG answer mini-template

Disease: brown_blight Severity: {severity_level} ({severity_percent}%) Advice: retrieve the brown_blight + {severity_level} action block. Add monitoring and safety note. Do not invent unsupported pesticide dosages.

Agent Card: gray_blight

Field	Value
Canonical class label	gray_blight
Human-readable name	Gray blight / Grey blight
Problem type	Fungal leaf disease
Retrieval aliases	gray blight, grey blight, Pestalotiopsis blight, tea gray leaf blight
Best query keywords	gray_blight treatment; grey blight tea moderate; Pestalotiopsis tea advice; gray_blight severe management

Recognition clues for the agent

- Gray to brown necrotic patches
- Lesions may have darker margins
- May show pale or grayish centers
- Favored by cool/wet/humid conditions and plant stress

Short explanation to include in answers

The model identified gray/grey blight-like lesions. Severity is based on pixel area of infected regions.

Severity-specific action blocks

Severity level	Agent treatment/advice block
mild (>0-15%)	Increase monitoring during wet periods Remove small affected leaves if practical Improve air movement through pruning and spacing Maintain balanced nutrition, especially potassium
moderate (>15-40%)	Remove infected material and reduce field humidity Avoid working in wet fields where spread can increase Use locally registered fungicide for tea blight diseases according to label Re-inspect after 7-14 days
severe (>40%)	Seek expert confirmation because severe gray blight can spread quickly Prune severely affected growth where appropriate Use a registered fungicide plan and rotate active ingredient groups Maintain strict sanitation of tools and removed material

Do-not-recommend / safety guardrails

- Do not leave infected debris in the field
- Do not prune with unclean tools
- Do not ignore repeated infection after spraying; reassess drainage and shade

RAG answer mini-template

Disease: gray_blight Severity: {severity_level} ({severity_percent}%) Advice: retrieve the gray_blight + {severity_level} action block. Add monitoring and safety note. Do not invent unsupported pesticide dosages.

Agent Card: algal_leaf_spot

Field	Value
Canonical class label	algal_leaf_spot
Human-readable name	Algal leaf spot
Problem type	Parasitic algal leaf disease
Retrieval aliases	algal leaf spot, Cephaleuros leaf spot, rust-red leaf spot, orange algal spots
Best query keywords	algal_leaf_spot treatment; Cephaleuros tea advice; algal leaf spot mild moderate severe; copper-based algal spot tea

Recognition clues for the agent

- Small circular raised spots
- Orange, rusty, reddish-brown, or velvety appearance
- Often on older or shaded leaves
- More common in humid, shaded, poorly aerated areas

Short explanation to include in answers

The model identified algal leaf spot-like symptoms. The affected area is estimated from infected pixels.

Severity-specific action blocks

Severity level	Agent treatment/advice block
mild (>0-15%)	Improve sunlight penetration and airflow through pruning Reduce excessive shade and moisture Remove heavily affected older leaves where practical Monitor after rain or humid periods
moderate (>15-40%)	Combine cultural control with a locally registered copper-based product if recommended for tea Correct drainage and shade problems Avoid prolonged wet leaf surface Recheck in 2-3 weeks because algal problems respond slowly
severe (>40%)	Consult expert/extension service to confirm algal infection Use registered copper-based control only as allowed locally Prune heavily affected shaded sections and improve field hygiene Monitor new growth for recurrence

Do-not-recommend / safety guardrails

- Do not treat algal leaf spot as an insect pest
- Do not rely only on spraying if shade and humidity remain high
- Avoid unnecessary pesticide use when cultural correction is enough

RAG answer mini-template

Disease: algal_leaf_spot Severity: {severity_level} ({severity_percent}%) Advice: retrieve the algal_leaf_spot + {severity_level} action block. Add monitoring and safety note. Do not invent unsupported pesticide dosages.

Agent Card: helopeltis

Field	Value
Canonical class label	helopeltis
Human-readable name	Helopeltis / Tea mosquito bug
Problem type	Insect pest injury, not fungal disease
Retrieval aliases	helopeltis, tea mosquito bug, Helopeltis theivora, TMB, mirid bug injury
Best query keywords	helopeltis treatment; tea mosquito bug moderate advice; Helopeltis severe IPM; helopeltis pest not fungus; tea mosquito bug scouting

Recognition clues for the agent

- Small puncture marks on young shoots or leaves
- Brown or black necrotic spots around feeding punctures
- Distorted, scorched, or blighted young shoots
- Damage can appear in clusters on tender flushes

Short explanation to include in answers

The model identified symptoms consistent with Helopeltis feeding injury. Severity reflects damaged pixels, but pest activity should be confirmed by field scouting.

Severity-specific action blocks

Severity level	Agent treatment/advice block
mild (>0-15%)	Scout young shoots and surrounding bushes for active bugs or fresh feeding spots Remove badly damaged shoots during plucking where practical Improve shade regulation and field sanitation Avoid unnecessary insecticide if damage is old and pest is not active
moderate (>15-40%)	Increase monitoring of tender flushes and nearby sections Remove alternate hosts/weeds around affected sections where recommended Use IPM first: sanitation, shade management, drainage, timely plucking If active pest pressure is confirmed, use locally registered selective insecticide only according to label and extension guidance
severe (>40%)	Consult tea pest specialist/extension service quickly Mark affected block and monitor edge rows because spread may start at field margins Consider targeted/barrier treatment only with locally registered products and PPE Recheck after 7-10 days and avoid repeated unsupervised spraying

Do-not-recommend / safety guardrails

- Do not recommend fungicide for Helopeltis; it is an insect pest
- Do not use banned products or unregistered insecticides
- Do not advise spray mixing or exact chemical recipes without local label guidance

RAG answer mini-template

Disease: helopeltis Severity: {severity_level} ({severity_percent}%) Advice: retrieve the helopeltis + {severity_level} action block. Add monitoring and safety note. Do not invent unsupported pesticide dosages.

Agent Card: healthy

Field	Value
Canonical class label	healthy
Human-readable name	Healthy leaf
Problem type	No visible disease class detected
Retrieval aliases	healthy, no disease, normal tea leaf, disease-free leaf
Best query keywords	healthy tea leaf advice; no disease tea maintenance; healthy class no treatment

Recognition clues for the agent

- Uniform green leaf appearance
- No clear necrotic lesions or pest injury
- No predicted infected region

Short explanation to include in answers

The model predicted healthy. Severity should be 0% or near 0%.

Severity-specific action blocks

Severity level	Agent treatment/advice block
mild (>0-15%)	Not applicable. If severity is above 0 but class is healthy, tell user to re-check image quality and model confidence.
moderate (>15-40%)	Not applicable. Ask for a clearer image or expert verification if model outputs conflict.
severe (>40%)	Not applicable. A healthy class with severe severity indicates pipeline conflict; request re-analysis.

Do-not-recommend / safety guardrails

- Do not recommend pesticide for healthy leaves
- Do not create disease advice if model confidence is low and image quality is poor

RAG answer mini-template

Disease: healthy Severity: {severity_level} ({severity_percent}%) Advice: retrieve the healthy + {severity_level} action block. Add monitoring and safety note. Do not invent unsupported pesticide dosages.

Conflict Handling Rules

Situation	Agent response
Predicted disease is healthy but severity > 0%	Mention inconsistency. Ask user to re-run image or inspect segmentation. Provide general monitoring advice only.
Predicted disease is helopeltis but retrieved fungal advice	Correct the response: Helopeltis is an insect pest injury. Do not recommend fungicide as the main solution.
Low confidence classification	Say the result is uncertain. Provide cautious general field hygiene advice and suggest expert confirmation.
Severe severity level	Give urgent escalation: expert consultation, sanitation/pruning, and locally registered treatment only.
User asks for exact pesticide recipe	Avoid exact mixing instructions. Say to follow product label, local registration, PPE, and pre-harvest interval.

RAG Retrieval Examples

Input from model	Best retrieval intent	Answer focus
predicted_disease=brown_blight; severity_level=mild	brown_blight mild advice	Remove infected leaves, reduce humidity, monitor, preventive label-based fungicide if risk increases.
predicted_disease=helopeltis; severity_level=moderate	helopeltis moderate IPM	Scout active pest, sanitation, shade management, remove alternate hosts/weeds, registered selective insecticide only if active pest confirmed.
predicted_disease=algal_leaf_spot; severity_level=severe	algal_leaf_spot severe copper shade	Confirm algal infection, improve shade/airflow/drainage, use registered copper-based product if locally recommended.
predicted_disease=healthy; severity_level=healthy	healthy maintenance	No treatment, monitoring, hygiene, nutrition, drainage.

Source and Evidence Notes

This knowledge base was adapted from the user-provided tea disease treatment guide and converted into agent-ready RAG chunks. The Helopeltis section uses an IPM framing: regular scouting, field sanitation, pruning/plucking management, shade regulation, alternate-host/weed management, and careful use of locally registered insecticides only when active pest pressure is confirmed.

For all chemical recommendations, the RAG agent must avoid banned or unregistered pesticides and must not invent dosages. It should recommend following local label instructions, BTRI/extension guidance, PPE requirements, and pre-harvest intervals.

Recommended citation text for thesis: A structured RAG knowledge base was prepared from tea disease and pest management information. Disease-specific cards were written using the same labels as the trained model, and each card included symptoms, severity-based actions, safety guardrails, and retrieval keywords to support grounded advisory generation.